

Weakly compact operators into separable BAP spaces factor through reflexive spaces with bases

Abstract

We prove that if Y is separable and has the bounded approximation property, then every weakly compact operator $T : X \rightarrow Y$ factors through a reflexive Banach space with a Schauder basis. In particular, if Y has a Schauder basis, then every weakly compact operator into Y factors through a reflexive Banach space with a Schauder basis. The proof combines a tail-convexification argument, a $c(Y)$ -encoding of finite-rank approximants, the Davis–Figiel–Johnson–Pełczyński interpolation construction, and Johnson’s reflexive basisification theorem.

1 Statement of the result

Throughout, Banach spaces are over $\mathbb{R}$ or $\mathbb{C}$. We write B_E for the closed unit ball of a Banach space E , and we identify E with its canonical image in E^{**} when no confusion can arise.

The main theorem is the following.

Theorem 1.1. *Let Y be a separable Banach space with the bounded approximation property. If $T : X \rightarrow Y$ is weakly compact, then T factors through a reflexive Banach space with a Schauder basis.*

Since every Banach space with a Schauder basis is separable and has the bounded approximation property, we immediately obtain the form relevant to the MathOverflow question.

Corollary 1.2. *Let Y have a Schauder basis. If $T : X \rightarrow Y$ is weakly compact, then T factors through a reflexive Banach space with a Schauder basis.*

The only external structural result used at the end is the following classical theorem of Johnson, the reflexive version of Pełczyński’s basisification theorem.

Theorem 1.3 (Johnson). *Every separable reflexive Banach space with the bounded approximation property is isomorphic to a complemented subspace of a reflexive Banach space with a Schauder basis.*

For reference, Pełczyński proved that a separable Banach space has the bounded approximation property if and only if it is isomorphic to a complemented subspace of a Banach space with a Schauder basis [3]; Johnson proved the reflexive strengthening in [2].

2 Tail convexification

We first isolate the approximation device needed in the proof.

Lemma 2.1. *Let Y be separable and have the bounded approximation property. Then there are finite-rank operators $P_n : Y \rightarrow Y$ such that*

$$\sup_n \|P_n\| < \infty \quad \text{and} \quad P_n y \rightarrow y \quad (y \in Y).$$

Proof. Let (y_j) be dense in Y . If Y has the λ -bounded approximation property, choose, for each n , a finite-rank operator P_n with $\|P_n\| \leq \lambda$ and

$$\|P_n y_j - y_j\| < 1/n \quad (1 \leq j \leq n).$$

Then $P_n y_j \rightarrow y_j$ for every j . Since the P_n are uniformly bounded and (y_j) is dense, $P_n y \rightarrow y$ for every $y \in Y$. $\square$

Lemma 2.2. *Let Y be separable and let $T : X \rightarrow Y$ be weakly compact. Put*

$$K = (B_{Y^*}, w^*)$$

and define

$$F : K \rightarrow X^*, \quad F(y^*) = T^* y^*.$$

Then F , regarded as an X^ -valued map with the norm topology on X^* , is Baire-one.*

Proof. Let

$$W = \overline{T(B_X)}^w \subset Y.$$

Since T is weakly compact, W is weakly compact. Since Y is separable, W is weakly compact metrizable. Choose a weakly dense sequence (w_j) in W .

For $y^*, z^* \in B_{Y^*}$,

$$\begin{aligned} \|T^* y^* - T^* z^*\| &= \sup_{x \in B_X} |(y^* - z^*)(Tx)| \\ &= \sup_{w \in W} |(y^* - z^*)(w)| \\ &= \sup_j |(y^* - z^*)(w_j)|. \end{aligned}$$

Thus $F(K)$ is isometric to a subset of $C(W, w)$, and therefore $F(K)$ is separable.

Moreover, if $z^* \in K$ and $r \geq 0$, then

$$\{y^* \in K : \|F(y^*) - F(z^*)\| \leq r\} = \bigcap_j \{y^* \in K : |(y^* - z^*)(w_j)| \leq r\},$$

which is weak-star closed in K . Hence the inverse image under F of every norm-open ball centered at a point of a countable dense subset of $F(K)$ is an F_σ subset of K . Since $F(K)$ is separable, it follows that the inverse image of every norm-open subset of X^* is an F_σ subset of K . By the Lebesgue–Hausdorff characterization of Baire-one maps into metric spaces, F is Baire-one. $\square$

Lemma 2.3 (Tail convexification). *Let K be compact metrizable, let E be a Banach space, and let $f_n : K \rightarrow E$ be uniformly bounded norm-continuous maps. Suppose that*

$$f_n(t) \xrightarrow{w} f(t) \quad (t \in K),$$

and that $f : K \rightarrow E$ is norm-Baire-one. Then there exist

$$g_k \in \text{conv}\{f_n : n \geq k\}$$

such that

$$\|g_k(t) - f(t)\| \rightarrow 0 \quad (t \in K).$$

Proof. Since f is bounded and Baire-one, choose uniformly bounded norm-continuous maps $u_n : K \rightarrow E$ such that $u_n(t) \rightarrow f(t)$ in norm for every $t \in K$. Indeed, start with any continuous approximating sequence and compose it with a radial retraction onto a closed ball containing $f(K)$.

Set $h_n = f_n - u_n$. Let

$$L = K \times (B_{E^*}, w^*).$$

For $(t, e^*) \in L$, define

$$\widehat{h}_n(t, e^*) = e^*(h_n(t)).$$

Each $\widehat{h}_n$ belongs to $C(L)$, the sequence $(\widehat{h}_n)$ is uniformly bounded, and $\widehat{h}_n(t, e^*) \rightarrow 0$ pointwise on L , because $f_n(t) \rightarrow f(t)$ weakly and $u_n(t) \rightarrow f(t)$ in norm. By dominated convergence against regular Borel measures on L , $\widehat{h}_n \rightarrow 0$ weakly in $C(L)$.

Fix k . Since 0 lies in the weak closure of $\text{conv}\{\widehat{h}_n : n \geq k\}$, Mazur's theorem gives an element of this convex hull with $C(L)$ -norm less than $1/k$. Thus there are finite convex coefficients $(\alpha_n^{(k)})_{n \geq k}$ such that

$$\left\| \sum_{n \geq k} \alpha_n^{(k)} \widehat{h}_n \right\|_{C(L)} < 1/k.$$

Define

$$g_k = \sum_{n \geq k} \alpha_n^{(k)} f_n, \quad v_k = \sum_{n \geq k} \alpha_n^{(k)} u_n.$$

Then

$$\sup_{t \in K} \|g_k(t) - v_k(t)\| < 1/k.$$

For each fixed $t \in K$, since $u_n(t) \rightarrow f(t)$ in norm, every convex combination of the tail $(u_n(t))_{n \geq k}$ tends to $f(t)$ as $k \rightarrow \infty$. Hence $v_k(t) \rightarrow f(t)$, and therefore $g_k(t) \rightarrow f(t)$ in norm. $\square$

Proposition 2.4. *Let Y be separable with the bounded approximation property, and let $T : X \rightarrow Y$ be weakly compact. Then there are finite-rank operators $A_k : X \rightarrow Y$ such that*

$$A_k^* y^* \rightarrow T^* y^* \quad (y^* \in Y^*), \tag{1}$$

$$A_k^{**} x^{**} \rightarrow T^{**} x^{**} \quad (x^{**} \in X^{**}), \tag{2}$$

in norm.

Proof. Choose finite-rank operators $P_n : Y \rightarrow Y$ as in Lemma 2.1. Put $K = (B_{Y^*}, w^*)$ and define

$$F_n : K \rightarrow X^*, \quad F_n(y^*) = T^*P_n^*y^*,$$

while $F(y^*) = T^*y^*$.

Each F_n is norm-continuous because $P_n^*Y^*$ is finite-dimensional. The sequence (F_n) is uniformly bounded. For $x^{**} \in X^{**}$ and $y^* \in K$,

$$\langle F_n(y^*) - F(y^*), x^{**} \rangle = \langle y^*, (P_n - I_Y)T^{**}x^{**} \rangle.$$

Since T is weakly compact, $T^{**}x^{**} \in Y$, and hence $P_n T^{**}x^{**} \rightarrow T^{**}x^{**}$ in norm. Thus

$$F_n(y^*) \xrightarrow{w} F(y^*) \quad (y^* \in K).$$

By Lemma 2.2, F is norm-Baire-one. Lemma 2.3 therefore yields convex combinations

$$S_k \in \text{conv}\{P_n : n \geq k\}$$

such that

$$\|T^*S_k^*y^* - T^*y^*\| \rightarrow 0 \quad (y^* \in B_{Y^*}).$$

By homogeneity the same convergence holds for every $y^* \in Y^*$.

Set

$$A_k = S_k T.$$

Then each A_k is finite-rank and (1) holds. Moreover, for $y \in Y$,

$$\|S_k y - y\| \leq \sup_{n \geq k} \|P_n y - y\| \rightarrow 0,$$

because S_k is a convex combination of the tail $\{P_n : n \geq k\}$. If $x^{**} \in X^{**}$, then $T^{**}x^{**} \in Y$, and therefore

$$A_k^{**}x^{**} = S_k^{**}T^{**}x^{**} = S_k T^{**}x^{**} \rightarrow T^{**}x^{**}.$$

This proves (2). □

3 The bridge theorem

The next result converts the two-sided finite-rank approximation from Proposition 2.4 into a factorization through a reflexive space with the bounded approximation property. The final basis is then supplied by Theorem 1.3.

Theorem 3.1 (Bridge theorem). *Let $T : X \rightarrow Y$ be an operator. Suppose there are finite-rank operators $A_n : X \rightarrow Y$ such that*

$$A_n^* y^* \rightarrow T^* y^* \quad (y^* \in Y^*), \tag{3}$$

$$A_n^{**} x^{**} \rightarrow T^{**} x^{**} \quad (x^{**} \in X^{**}), \tag{4}$$

in norm. Then T factors through a reflexive Banach space with a Schauder basis.

Proof. By uniform boundedness, after increasing the constant if necessary, fix

$$M \geq \max\{\|T\|, \sup_n \|A_n\|\} < \infty.$$

Condition (4), applied to the canonical copy of X in X^{**} , implies that $A_n x \rightarrow Tx$ for every $x \in X$.

Let

$$E = c(Y) = \{(y_n) : y_n \text{ converges in norm in } Y\}$$

with its supremum norm. Define

$$R : X \rightarrow E, \quad Rx = (A_n x)_{n=1}^\infty,$$

and let

$$L : E \rightarrow Y, \quad L((y_n)) = \lim_n y_n.$$

Then $T = LR$.

We first prove that R is weakly compact. Every functional $\Phi \in E^*$ admits a representation

$$\Phi((z_n)) = \eta^*\left(\lim_n z_n\right) + \sum_{n=1}^\infty y_n^*(z_n), \quad (5)$$

where $\eta^* \in Y^*$ and $(y_n^*) \in \ell_1(Y^*)$. This follows from the isomorphism

$$c(Y) \cong c_0(Y) \oplus_\infty Y, \quad (z_n) \mapsto ((z_n - \lim z_n)_n, \lim z_n).$$

For $x^{**} \in X^{**}$, condition (4) gives a convergent sequence

$$\hat{R}x^{**} := (A_n^{**}x^{**})_{n=1}^\infty \in E,$$

with limit $T^{**}x^{**}$. Using (5),

$$\Phi(\hat{R}x^{**}) = \eta^*(T^{**}x^{**}) + \sum_{n=1}^\infty y_n^*(A_n^{**}x^{**}) = x^{**}(R^*\Phi).$$

Thus $R^{**}x^{**}$ belongs to the canonical copy of E in E^{**} for every $x^{**} \in X^{**}$. Hence R is weakly compact.

For $m \in \mathbb{N}$, define the freezing projection

$$Q_m : E \rightarrow E, \quad Q_m(y_1, y_2, \dots) = (y_1, \dots, y_m, y_m, y_m, \dots).$$

Then

$$\|Q_m\| \leq 1, \quad Q_m Q_k = Q_{\min(m,k)}, \quad Q_m z \rightarrow z \quad (z \in E). \quad (6)$$

We next record the key shrinking estimate. Let $\Phi \in E^*$ be represented as in (5). Since

$$(I - Q_m)Rx = (0, \dots, 0, (A_{m+1} - A_m)x, (A_{m+2} - A_m)x, \dots)$$

and this sequence has limit $(T - A_m)x$, we have

$$R^*(I - Q_m^*)\Phi = (T^* - A_m^*)\eta^* + \sum_{n>m} (A_n^* - A_m^*)y_n^*.$$

Consequently,

$$\|R^*(I - Q_m^*)\Phi\| \leq \|(T^* - A_m^*)\eta^*\| + 2M \sum_{n>m} \|y_n^*\| \rightarrow 0.$$

Thus

$$\|R^*(I - Q_m^*)\Phi\| \rightarrow 0 \quad (\Phi \in E^*). \quad (7)$$

Let

$$W = \overline{R(B_X)}^w \subset E.$$

Since R is weakly compact, W is weakly compact. Define

$$\mathcal{K} = \overline{\text{aco}} \left(\bigcup_{m=1}^{\infty} Q_m W \right), \quad (8)$$

where the closure is norm closure and aco denotes the absolutely convex hull.

We claim that $\mathcal{K}$ is weakly compact. It is enough to show that $\bigcup_m Q_m W$ is relatively weakly compact. Take a sequence $Q_{m_j} w_j$ with $w_j \in W$. Passing to a subsequence, either (m_j) is constant or $m_j \rightarrow \infty$. The first case follows from weak compactness of $Q_m W$. In the second case, pass to a further subsequence with $w_j \xrightarrow{w} w \in W$. For every $\Phi \in E^*$, using (7),

$$\begin{aligned} \sup_{v \in W} |\Phi(Q_m v - v)| &= \sup_{x \in B_X} |\Phi(Q_m R x - R x)| \\ &= \|R^*(Q_m^* - I)\Phi\| \rightarrow 0. \end{aligned}$$

Hence $\Phi(Q_{m_j} w_j - w_j) \rightarrow 0$, and therefore $Q_{m_j} w_j \xrightarrow{w} w$. By Eberlein–Šmulian, $\bigcup_m Q_m W$ is relatively weakly compact. Krein's theorem then implies that the closed absolutely convex hull $\mathcal{K}$ is weakly compact.

Moreover,

$$W \subset \mathcal{K}, \quad (9)$$

because $Q_m w \rightarrow w$ in norm for every $w \in W$, and

$$Q_m \mathcal{K} \subset \mathcal{K} \quad (m \in \mathbb{N}), \quad (10)$$

because $Q_m Q_k = Q_{\min(m,k)}$.

We now apply the Davis–Figiel–Johnson–Pełczyński interpolation construction to $\mathcal{K}$. For $j \in \mathbb{Z}$, set

$$U_j = 2^j \mathcal{K} + 2^{-j} B_E,$$

and let $|\cdot|_j$ be the Minkowski functional of U_j . Define

$$Z = \left\{ z \in E : \|z\|_Z^2 := \sum_{j \in \mathbb{Z}} |z|_j^2 < \infty \right\}. \quad (11)$$

The standard DFJP theorem says that Z is reflexive, the inclusion $J : Z \hookrightarrow E$ is bounded, and $\mathcal{K}$ is bounded as a subset of Z ; see [1]. Since $R(B_X) \subset W \subset \mathcal{K}$, the map $R : X \rightarrow E$ lifts to a bounded operator

$$\tilde{R} : X \rightarrow Z, \quad J\tilde{R} = R.$$

Therefore

$$T = LJ\tilde{R}.$$

So T already factors through the reflexive space Z .

It remains to prove that Z has the bounded approximation property. By (10) and $\|Q_m\| \leq 1$, one has $Q_m U_j \subset U_j$ for every $j \in \mathbb{Z}$. Hence each Q_m restricts to a contraction on Z . Also $Q_m z \rightarrow z$ in Z for every $z \in Z$. Indeed, for fixed j ,

$$|(I - Q_m)z|_j \leq 2^j \|(I - Q_m)z\|_E \rightarrow 0,$$

while

$$|(I - Q_m)z|_j \leq |z|_j + |Q_m z|_j \leq 2|z|_j.$$

Dominated convergence in the $\ell_2(\mathbb{Z})$ -sum gives

$$\|Q_m z - z\|_Z \rightarrow 0 \quad (z \in Z). \quad (12)$$

We show that $Q_m|_Z$ is finite-rank. Let

$$H = \overline{\text{span}}^E \mathcal{K}.$$

First, $Z \subset H$. If $q : E \rightarrow E/H$ is the quotient map, then $q(\mathcal{K}) = 0$, and so

$$q(U_j) \subset 2^{-j} B_{E/H}.$$

Consequently,

$$|z|_j \geq 2^j \|qz\| \quad (j \geq 0).$$

If $qz \neq 0$, then $\sum_{j \geq 0} |z|_j^2 = \infty$. Hence every $z \in Z$ lies in H .

Fix m and set

$$F_m = \sum_{i=1}^m Q_i R(X).$$

Each $Q_i R(X)$ is finite-dimensional, since

$$Q_i R x = (A_1 x, \dots, A_i x, A_i x, A_i x, \dots)$$

and each A_k is finite-rank. Thus F_m is finite-dimensional. Since $Q_i R(X)$ is finite-dimensional and hence weakly closed, $Q_i W \subset Q_i R(X)$. Therefore, for every generator $Q_k w$ of the set in (8),

$$Q_m Q_k w = Q_{\min(m,k)} w \in F_m.$$

By convexity and norm closure,

$$Q_m \mathcal{K} \subset F_m.$$

Since $Z \subset H = \overline{\text{span}}^E \mathcal{K}$, it follows that

$$Q_m Z \subset F_m.$$

Thus $Q_m|_Z$ is finite-rank. Together with (12), this shows that Z has the metric approximation property. It is separable as well, because the increasing union of the finite-dimensional spaces $Q_m Z$ is dense in Z .

Now apply Theorem 1.3. There is a reflexive Banach space V with a Schauder basis such that Z is isomorphic to a complemented subspace of V . Let $i : Z \rightarrow V$ be an embedding and let $P : V \rightarrow iZ$ be a bounded projection. Define

$$B : X \rightarrow V, \quad B = i\tilde{R},$$

and

$$C : V \rightarrow Y, \quad C = (LJ)i^{-1}P.$$

Then $CB = T$. Therefore T factors through the reflexive basis space V . $\square$

4 Proof of the main theorem

Proof of Theorem 1.1. By Proposition 2.4, there are finite-rank operators $A_n : X \rightarrow Y$ satisfying (3) and (4). The bridge theorem, Theorem 3.1, then gives a factorization of T through a reflexive Banach space with a Schauder basis. $\square$

Remark 4.1. *The proof actually constructs, before the final basisification step, a separable reflexive DFJP interpolation space Z with the metric approximation property through which T factors. Johnson's theorem is used only to replace this reflexive MAP space by a complemented subspace of a reflexive space with a genuine Schauder basis.*

References

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